

DeXposure-Claw: An Agentic System for DeFi Risk Monitoring and Decision Recommendation Based on a Graph Time-Series Foundation Model

Aijie Shu and Fengxiang He

Abstract—Decentralised finance (DeFi) has become a densely interconnected financial system in which credit exposures propagate through tokenised liabilities, making timely *risk monitoring* and *decision recommendation* essential for investors, protocol operators, and regulators. Existing agentic pipelines for DeFi risk lack a dedicated forecasting backbone for graph-structured time series, limiting their data efficiency, uncertainty calibration, and robustness. Building on DeXposure-FM, a graph time-series foundation model pre-trained on 43.7 million exposure-graph entries, we present *DeXposure-Claw*: a four-layer agentic architecture—FM prediction, deterministic monitor and scenario engines, LLM decision, and a data-health/confidence safety gate—that turns predicted exposure graphs into auditable supervisory tickets. We also introduce *DeXposure-Bench*, a six-axis evaluation suite for FM + LLM decision pipelines on temporal exposure graphs, with a regulator-aligned absolute-loss ground truth and eight reference method implementations. On the frozen 2025 decision split, the full FM + LLM + gate stack raises end-to-end ticket F1 from 0.0076 (persistence + rules) to 0.0233 (+208%) and LLM-as-judge explanation quality from 2.03 to 2.69/5 (+33%). The forecasting layer does not uniformly dominate persistence on static point metrics; its value appears in directional signal, calibrated prediction intervals (coverage 0.913 against a 0.90 target, ECE 0.013), and lower mean degradation under corruption (0.113 vs. 0.148). In a separate historical early-warning event study, the shared weighted-degree monitor gives 4–5 week lead times on Terra/Luna, FTX, and SVB-USDC. These gains are a trade-off rather than uniform dominance: FM + LLM variants improve recall and explanations at a non-trivial false-intervention cost ($FIR \approx 0.44$) compared with rule-based baselines ($FIR = 0$). We release the harness, dataset, reference implementations, and ticket-level audit logs to support extension.

Index Terms—Agentic system, graph, time series, foundation model, decentralised finance, financial network, risk modelling.



1 INTRODUCTION

In May 2022, the Terra/Luna collapse erased over \$40 billion in value within 48 hours, triggering cascading liquidations across lending protocols, DEXs, and yield aggregators that held UST-denominated assets. Five months later, the FTX insolvency exposed hidden cross-protocol exposures and wiped out another \$8 billion. These crises share a common pattern: shocks propagated through an interconnected web of tokenised liabilities—credit exposures—that no single participant fully observed. In decentralised finance, exposures are high-dimensional, time-varying, and often implicit; data are noisy and incomplete; and crises unfold within days, leaving limited time for manual assessment.

Automated *agentic* pipelines offer a principled response: an agent can ingest on-chain measurements, maintain a belief over the evolving exposure network, surface early-warning signals, stress-test plausible shocks, and recommend risk-mitigating actions with auditable rationale. However, existing agentic proposals for DeFi risk largely rely on general-purpose large language models or static heuristics, and lack a dedicated forecasting backbone for graph-structured time series. Without accurate and data-efficient

prediction of network dynamics, monitoring agents suffer from high false-alarm rates and delayed detection, while decision agents risk recommending actions based on brittle or poorly calibrated beliefs. In high-stakes contexts, uncertainty calibration and robustness to missingness and regime shifts are not optional—they determine whether an agent is trusted and deployable.

This paper addresses that bottleneck by building an agentic system around a strong graph time-series foundation model, and—in tandem—by providing the evaluation suite the community needs to compare such systems. We leverage DEXPOSURE-FM, a foundation model trained on over 43.7 million data entries from DeFi exposure networks, designed to forecast the evolution of credit-exposure graphs across multiple horizons. On top of this backbone we present DEXPOSURE-CLAW, a modular four-layer query architecture that follows a *forecast–measure–decide* paradigm, and DEXPOSURE-BENCH, a standardised evaluation suite that benchmarks FM+LLM decision pipelines on a shared regulator-aligned ground truth.

Our contributions are:

- 1) **DEXPOSURE-CLAW: a four-layer agentic architecture for DeFi exposure-network risk (§4).** We design and implement an end-to-end system organised as four layers—FM prediction, deterministic monitor and scenario engines, LLM decision, and a safety gate—that transforms

- A. Shu and F. He are with the School of Informatics, University of Edinburgh, Edinburgh EH8 9AB, U.K. (e-mail: v1ashu@ed.ac.uk; F.He@ed.ac.uk).
- Corresponding author: Fengxiang He.

forecasted exposure graphs into auditable supervisory tickets. The monitor, scenario, and safety-gate layers are deterministic and reproducible; the LLM decision layer is governed by per-ticket audit logs and exhibits 0.78–0.86 inter-run Jaccard consistency. Five network risk metrics with rolling baselines, five standardised stress scenarios with CVaR aggregation, and a four-level playbook gated by data-health and confidence checks complete the design. While prior agentic work in DeFi targets transaction verification [1] or intent mining [2], and graph-based foundation models for risk have been explored only in traditional finance [3], no existing system unifies these capabilities for DeFi exposure networks.

- 2) **Empirical effectiveness of the FM+LLM agentic stack with quantified Pareto trade-offs (§6).** On DEXPOSURE-BENCH we show that DEXPOSURE-CLAW (FM+LLM+gate) raises end-to-end ticket F1 from 0.0076 (persistence+rules) to 0.0233 (+208%) and LLM-as-judge explanation quality from 2.03 to 2.69/5 (+33%). The forecasting layer is useful for monitoring rather than uniformly better on every point metric: persistence remains stronger on PageRank MAE and static rank correlation, while the FM supplies directional trend signal (TrendCons@h4 0.628 vs a structurally pinned 0 for persistence), calibrated prediction intervals (coverage 0.913 vs 0.90 target, ECE 0.013), and lower mean degradation under corruption (0.113 vs 0.148). A separate historical b2_warning event study shows 4–5 week lead times for Terra/Luna, FTX, and SVB-USDC under the shared weighted-degree monitor; we do not treat that result as predictor-specific FM evidence. Layer-wise attribution against two distinct baselines makes the decision gains decomposable: adding the FM signal to a snapshot LLM (m2_snapshot_llm→m6_fm_llm) lifts F1 by +32%; replacing rules with an LLM under a fixed FM predictor (m5_fm_rules→m6_fm_llm) lifts F1 by another +27%. The bench also exposes the cost: FM+LLM variants raise recall and explanation at a non-trivial false-intervention rate (FIR $\approx$ 0.44 vs 0 for rule-based baselines), while the safety gate trades 3% F1 for +0.21 judge score and +10% run-to-run consistency.

- 3) **DEXPOSURE-BENCH: a reusable testbed for FM+LLM decision pipelines on temporal exposure graphs (§5).** We release a standardised harness that scores a candidate pipeline on six axes—forecasting quality, streaming anomaly detection, predictive uncertainty, what-if scenario fidelity, supervisory decision quality, and data-quality robustness. The bench operates on the full DEXPOSURE corpus (4,300 protocols, 24,300 tokens, 43.7M exposure entries, 283 weekly snapshots from 2020 to 2025 [4]), uses a regulator-aligned ground truth defined by *absolute* weight loss rather than fractional change, and ships eight reference method implementations covering persistence, EvolveGCN, snapshot-LLM, FM-only, FM+rules, and three FM+LLM configurations. Generic LLM-agent benchmarks (SWE-BENCH, AGENT-BENCH, HELM) do not evaluate domain-grounded decisions, and temporal-graph benchmarks (TGB, OGB) stop at structural prediction; DEXPOSURE-BENCH fills this benchmark gap for agentic risk pipelines.

We will release the implementation and benchmark artifacts with the project repository.

2 RELATED WORKS

This section positions DEXPOSURE-CLAW against five neighbouring threads: financial-network contagion, temporal graph learning, foundation models for time series and graphs, LLM agents in finance and DeFi, and benchmark design.

2.1 Financial Network Risk and DeFi Contagion

Systemic-risk analysis in traditional finance has long treated the financial system as a network of balance-sheet obligations, funding links, and correlated asset exposures. Clearing-network models formalise how defaults propagate through interbank liabilities [5], while subsequent work studies when dense connections stabilise the system and when they amplify shocks [6], [7]. DebtRank-style measures further emphasise that a node can be “too central to fail” even when it has not yet defaulted [8]. These models motivate our focus on network position, concentration, and stress-test loss rather than protocol-level indicators alone.

DeFi changes the object being measured. Smart contracts replace part of the legal and operational layer, but composability, shared collateral, stablecoins, bridges, and liquidations create new contagion channels [9], [10], [11]. Recent work measures DeFi risk from aggregate deposits and borrowings [12], maps microscopic and systemic mechanisms across TradFi and DeFi [13], and studies fragility in TVL dynamics [14]. In contrast, our setting is an inter-protocol credit-exposure network: the units of analysis are weighted links between protocols, and the downstream task is not only to detect stress but also to produce traceable supervisory tickets.

2.2 Temporal Graph Learning for Dynamic Risk

Graph neural networks have become a natural tool for financial networks because they use both node attributes and relational structure. Prior work shows that GNNs can improve systemic-importance classification and systemic risk computation in simulated or interbank-style financial networks [15], [16], building on general graph convolution and attention architectures [17], [18]. These studies support the premise that risk is partly encoded in graph topology, not just in independent node features.

Our problem is additionally temporal: exposure weights, active protocols, and contagion paths evolve week by week. Dynamic-graph models such as DySAT, TGAT, EvolveGCN, and Temporal Graph Networks model changing neighbourhoods, temporal attention, recurrent graph filters, or event streams [19], [20], [21], [22]. Temporal graph benchmarks standardise link-prediction and node-classification tasks on dynamic graphs [23]. However, these settings typically optimise task-specific structural prediction. DEXPOSURE-CLAW instead uses temporal graph forecasts as inputs to monitoring, scenario stress tests, and LLM-based decisions; the forecasting layer is therefore evaluated by its effect on calibrated risk signals and recommendation tickets, not only by edge-level accuracy.

2.3 Foundation Models for Time Series and Graphs

Foundation models aim to replace narrowly trained task models with reusable representations learned from broad pretraining data [24]. Earlier neural forecasting architectures such as Temporal Fusion Transformers, Informer, and Autoformer established multi-horizon attention-based forecasting for long and heterogeneous time series [25], [26], [27]. More recently, Chronos, Lag-Llama, and TimesFM show that large pretrained sequence models can transfer across forecasting tasks and provide probabilistic forecasts [28], [29], [30]. In finance and economics, LLMs have also been used to extract signals from news and policy text for nowcasting pipelines [31], [32], [33].

Graph foundation modelling is less settled. Graphormer demonstrates that transformers can be made graph-aware through structural encodings [34]; self-supervised graph representation learning methods such as Deep Graph Infomax, GraphCL, and GraphMAE pretrain encoders through mutual information, contrastive augmentations, or masked attribute reconstruction [35], [36], [37]; and GraphPFN and TabPFN-style work explores how pretrained tabular predictors can transfer to graph-structured tasks [38], [39]. Heterogeneous graph pretraining has also been applied to default-risk propagation among bond issuers [3]. These systems motivate a foundation-model backbone for risk forecasting, but they do not directly address weighted temporal DeFi exposure graphs or the downstream requirements of uncertainty-gated supervisory decision making.

2.4 LLM Agents in Finance and DeFi

LLM agents are increasingly used for financial analysis, trading, and blockchain security. General agent patterns such as ReAct combine reasoning traces with tool-using actions [40], while financial LLM work such as FinGPT adapts language models to financial data and tasks [41]. In crypto portfolio management, multi-agent LLM systems coordinate market analysis and allocation decisions [42], and BlindTrade combines graph encoders with LLM agents for anonymised stock portfolio optimisation [43]. These systems are closest to ours in combining graph representations with language-model reasoning, but their objective is portfolio construction rather than systemic-risk supervision.

Within DeFi, agentic systems have focused mainly on security auditing, transaction understanding, and manipulation detection. Arbiter uses Graph-of-Thoughts reasoning for intent-transaction alignment [1]; Know Your Intent mines transaction intent with a multi-perspective LLM agent [2]; Heimdalr and Knowdit target smart-contract vulnerability detection [44], [45]; and PMDetector applies LLM reasoning to price manipulation detection [46]. Watson et al. combine graph anomaly detection with LLM-generated explanations for Bitcoin fraud [47]. These works show that LLMs can help explain or audit crypto systems, but they do not couple the agent with a dedicated graph time-series forecaster for network-level exposure risk.

2.5 Benchmarks for Graph, Forecasting, and Agentic Systems

The benchmark literature provides useful pieces but not the complete evaluation target required here. OGB and

TGB standardise graph and temporal graph evaluation [23], [48]; time-series forecasting work emphasises temporally valid splits and robust model selection [49]; and general LLM-agent benchmarks such as HELM, SWE-bench, and AgentBench evaluate language capabilities, software repair, or generic agent behaviour [50], [51], [52].

DEXPOSURE-BENCH occupies a different point in this space. It evaluates forecasting, early warning, calibration, scenario stress fidelity, decision quality, and data-quality robustness in one harness, with ground truth tied to absolute exposure loss. This lets us ask whether a foundation-model forecast actually improves downstream risk monitoring and ticket generation, and also whether safety gates reduce harmful recommendations under missingness or weak evidence.

3 NOTATION AND PRELIMINARIES

This section introduces the key terminology and notation used throughout the paper.

3.1 Credit Exposure in DeFi Networks

Let $X_G(q)$ be the set of tokens issued (or generated) by protocol q , and let $X_p(t)$ be the set of tokens held (or locked) by protocol p at time t . We say that protocol p has *credit exposure* to protocol q at time t if p holds at least one token issued by q :

$$X_p(t) \cap X_G(q) \neq \emptyset. \quad (1)$$

Equivalently, p is exposed to liabilities of q : if tokens issued by q devalue or become illiquid, p incurs losses. This token-mediated dependence generalises counterparty credit risk from traditional finance to DeFi, where exposures emerge through tokenised claims rather than explicit bilateral contracts. In practice, these dynamics can be tracked by reconstructing protocol-level TVL time series and monitoring how token composition changes over time [4].

Figure 1 shows how a single workflow can generate layered exposures. A user deposits ETH into Lido and receives stETH; stETH is wrapped into wstETH; wstETH is then deposited into Pendle, which splits it into PT-wstETH and YT-wstETH. On balance sheets, each step turns the upstream token into the downstream protocol's asset, so exposures stack: Pendle is exposed to the wstETH wrapper, which is exposed to Lido. If Lido suffers slashing or stETH de-pegs from ETH, losses can propagate through wstETH into Pendle and its PT/YT holders. This illustrates why DeFi exposures are naturally multi-layered and why DeXposure aims to measure them.

3.2 Mapping Tokens to Issuing Protocols

To implement Eq. (1), we must identify the issuing (or managing) protocol for each token. Let $\mathcal{P}$ be the set of protocols and $\mathcal{C}$ the set of blockchains. For any token σ appearing in protocol p , a *mapping function* $M(\sigma)$ returns the protocol q that issues or governs σ . We construct M via the following fallback pipeline:

- 1) *Metadata lookup*: match tokens to issuing protocols using DefiLlama metadata when available [53].

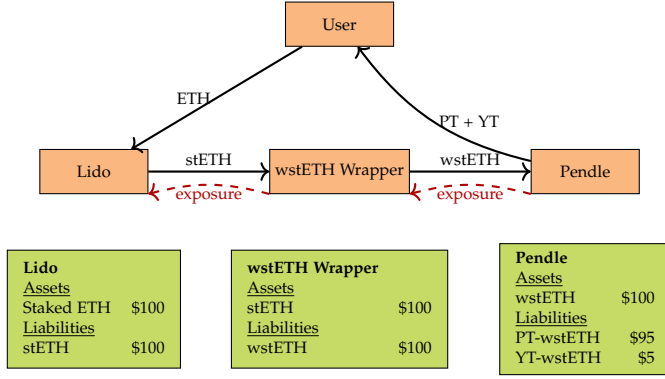


Fig. 1. Multi-layer credit exposure in DeFi. A user stakes ETH with Lido, receiving stETH, which is wrapped into wstETH and deposited into Pendle. Each protocol holds the liability of the preceding one, creating a chain of credit exposures from Pendle to Lido.

- 2) *Manual mapping*: for high-TVL tokens without metadata links, assign issuers using documentation and expert judgment.
- 3) *Text-vector similarity*: otherwise, vectorise token and protocol descriptions using TF-IDF [54] and assign $M(\sigma)$ as the most similar protocol, provided the cosine similarity exceeds a threshold θ .
- 4) *Primary market tokens*: if no match is obtained, treat the token as its own protocol (e.g., WETH).

This procedure attributes token movements to the appropriate issuer even when tokens are bridged or wrapped, and is used in curating DeXposure.

3.3 Network Representation and Value Flows

Over a discrete interval $\tau = t_2 - t_1$, we represent exposures by a weighted directed graph $G_\tau = (P_\tau, E_\tau)$, where P_τ is the set of active protocols and E_τ the directed edge set. Each node $p \in P_\tau$ receives a *node weight* equal to the value of its held assets at t_2 :

$$w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2, \sigma}, \quad (2)$$

where $v_{t_2, \sigma}$ is the USD value of token σ at time t_2 . Nodes below a threshold θ may be pruned.

To define the *edge weight* from p to q , we first compute token-level value flows:

$$F_{pq, \tau}^\sigma = \begin{cases} \max(0, -\Delta S_{p, \tau}^\sigma), & \text{if } \Delta S_{p, \tau}^\sigma < 0, \\ \max(0, \Delta S_{q, \tau}^\sigma), & \text{if } \Delta S_{q, \tau}^\sigma \geq 0, \end{cases} \quad (3)$$

where $\Delta S_{p, \tau}^\sigma = v_{p, t_2}^\sigma - v_{p, t_1}^\sigma$ is the change in the USD value of token σ held by p over τ . Intuitively, if p reduces holdings while q increases holdings, σ is interpreted as flowing from p to q ; the construction enforces non-negativity.

The *edge weight* $w_\tau(e_{pq})$ aggregates flows over tokens mapped to issuer q :

$$w_\tau(e_{pq}) = \sum_{\sigma \in X_{pq, \tau}} F_{pq, \tau}^\sigma, \quad (4)$$

where $X_{pq, \tau}$ is the set of tokens issued by q that move from p to q over τ . Positive $w_\tau(e_{pq})$ corresponds to increasing exposure from p to q ; for single-token protocols, that token's flow equals the edge weight. In summary, we compute

G_τ by aggregating holdings at t_1 and t_2 , assigning node weights, pruning small nodes, and summing positive token flows into issuer-specific edge weights. The resulting graph sequence captures the evolving credit-dependency structure curated in DeXposure.

4 ALGORITHM

This section specifies the agentic system built on DEXPOSURE-FM for two tasks: network risk monitoring (Section 4.3–4.4) and decision recommendation (Section 4.5).

Figure 2 provides an architectural overview of the full system.

4.1 System Interface

At each decision epoch t , the agent observes a credit-exposure snapshot $G_t = (V, E_t, W_t)$ and node covariates X_t , and interacts with DEXPOSURE-FM through a single forecast primitive:

$$\mathcal{P}_{t, h} \leftarrow \text{DEXPOSURE-FM.Forecast}(G_t, X_t, h), \quad \forall h \in \mathcal{H}, \quad (5)$$

where $\mathcal{P}_{t, h}$ denotes the model's predictive distribution over the future network G_{t+h} , and $\mathcal{H} = \{1, 4, 8, 12\}$ (weeks). From $\mathcal{P}_{t, h}$ we construct a representative predicted graph $\hat{G}_{t+h}$ using a thresholded expected-weight operator:

$$\hat{w}_{pq, t+h} := \Pr(e_{pq, t+h} = 1) \cdot \mathbb{E}[w_{pq, t+h} \mid e_{pq, t+h} = 1], \quad (6)$$

$$\hat{E}_{t+h} := \{(p, q) : \Pr(e_{pq, t+h} = 1) > \pi_{\min}\}. \quad (7)$$

where $\pi_{\min}$ (default 0.2, tuned on validation) balances edge coverage and sparsity. For uncertainty quantification, we draw M Monte Carlo samples $\{\tilde{G}_{t+h}^{(m)}\}_{m=1}^M$ by applying Bernoulli edge sampling and Gaussian weight perturbation to $\mathcal{P}_{t, h}$ (default $M = 50$).

The agent returns three outputs: (a) a set of monitoring alerts with evidence and confidence, (b) a scenario stress summary, and (c) a ranked list of recommendation tickets.

4.2 Data-Health Gating

Real-world DeFi graphs can be sparse, stale, or corrupted. Before producing actionable outputs, the agent computes a scalar data-health score from four deterministic checks—freshness, feature missingness, topology sanity (edge-to-node ratio), and discontinuity detection:

$$\begin{aligned} DH_t &:= \text{mean}(\text{fresh}_t, \text{miss}_t, \text{topo}_t, \text{disc}_t) \in [0, 1], \\ \text{SAFE_MODE}_t &:= \mathbb{I}\{DH_t < \tau_{\text{data}}\}. \end{aligned} \quad (8)$$

When DH_t falls below τ_{data} (default 0.7), the system enters safe mode: monitoring alerts are still emitted, but intervention-type recommendations are suppressed (Section 4.5).

4.3 Task I: Network Risk Monitoring

Monitoring metrics (Φ): Risk in exposure networks manifests as concentration (a few protocols dominate), fragility (shocks propagate widely), and instability (structure shifts rapidly). To capture these dimensions, the monitoring agent evaluates five functionals on $\hat{G}_{t+h}$:

Formally, at each horizon h we compute:

$$\mathbf{N}_{t, h} := \Phi(\hat{G}_{t+h}, X_t) \in \mathbb{R}^5, \quad \tilde{\mathbf{N}}_{t, h}^{(m)} := \Phi(\tilde{G}_{t+h}^{(m)}, X_t). \quad (9)$$

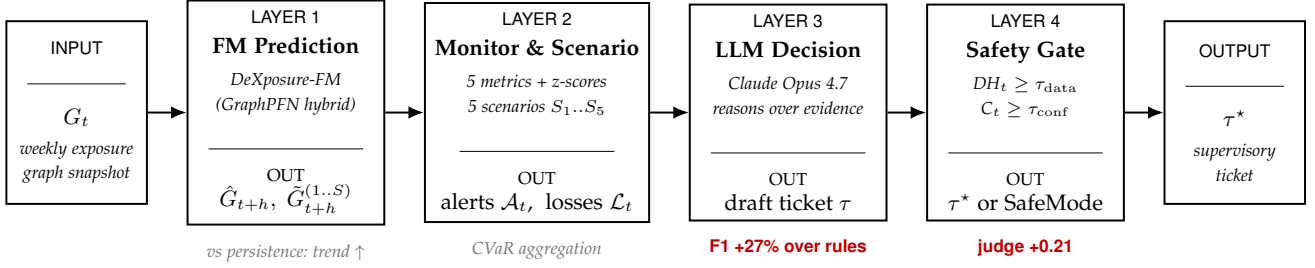


Fig. 2. DEXPOSURE-CLAW: a four-layer query architecture. The system ingests the weekly exposure graph G_t , runs FM prediction (Layer 1), monitoring and scenario engines (Layer 2), an LLM decision layer (Layer 3), and a safety gate (Layer 4) that emits either the supervisory ticket τ^* or a safe-mode notice.

ID	Metric	Definition
N1	Systemic Importance	$\max_{p \in V} \text{PageRank}(\hat{G}_{t+h})_p$
N2	HHI Concentration	$\sum_p (d_p / \sum_q d_q)^2$, d_p = weighted out-degree
N3	Network Density	$ E_{t+h} / V (V - 1)$
N4	PageRank Gini	$\text{Gini}(\{\text{PR}_p\}_{p \in V})$
N5	Degree Gini	$\text{Gini}(\{d_p\}_{p \in V})$

TABLE 1

Monitoring functionals Φ . PageRank uses power iteration with damping 0.85 and 10 iterations. Gini coefficient: $1 - 2 \sum_{i=1}^n \sum_{k=1}^i x_{(k)} / (n \cdot \sum_k x_{(k)})$, where $x_{(k)}$ are sorted values.

Alert generation: An alert fires when a predicted metric departs from its rolling historical baseline by more than z standard deviations (default $z = 2$, baseline window $L = 26$ weeks):

$$\mathcal{A}_t := \left\{ a = (h, j) : \frac{|N_{t,h}^{(j)} - \mu_t^{(j)}|}{\max(\sigma_t^{(j)}, \epsilon)} > z \right\}, \quad (10)$$

where $(\mu_t^{(j)}, \sigma_t^{(j)})$ are the rolling mean and standard deviation of metric j over the last L observed snapshots, and $\epsilon = 10^{-8}$ prevents division by zero.

Evidence and attribution: Each alert $a \in \mathcal{A}_t$ carries an evidence bundle $\text{Ev}_t(a) = (N_{t,h}^{(j)}, \mu_t^{(j)}, \sigma_t^{(j)}, \text{Attr}_t(a), h)$, where $\text{Attr}_t(a)$ is a ranked list of top- K contributing nodes/edges obtained via marginal contribution approximation: $\text{Contrib}(e) := \Phi^{(j)}(\hat{G}_{t+h}) - \Phi^{(j)}(\hat{G}_{t+h} \setminus e)$ (default $K = 10$).

Uncertainty-aware confidence: Alert reliability depends on three factors: data quality, forecast dispersion, and horizon length. Using Monte Carlo samples, we compute the coefficient of variation $\text{CV}_{t,h}^{(j)} := \sigma(\{\tilde{N}_{t,h}^{(m,j)}\}_m) / |\tilde{N}_{t,h}^{(j)}|$ (clamped to $[0, 1]$) and define:

$$C_t(a) := DH_t \cdot \frac{1}{1 + \text{CV}_{t,h}^{(j)}} \cdot \frac{1}{1 + \ln(1 + h)} \in [0, 1]. \quad (11)$$

This score monotonically decreases with lower data health, wider predictive spread, and longer horizons, ensuring that unreliable alerts are automatically downweighted.

4.4 Scenario Engine

To quantify tail risk beyond metric-level alerts, the agent evaluates a standardised stress-scenario library $\mathcal{S}_0$. Each scenario applies a deterministic shock—removing or reducing edge weights of targeted nodes—and propagates losses through the network:

ID	Scenario	Target	Shock
S1	Single protocol failure	Highest-weight node	100%
S2	Bridge cluster failure	All bridge-category nodes	100%
S3	Stablecoin de-peg	All stablecoin nodes	50%
S4	Sector-wide shock	All lending nodes	30%
S5	Correlated stress	Top-10 nodes by weight	20%

TABLE 2

Standardised stress-scenario library $\mathcal{S}_0$.

For each scenario s and horizon h , we compute the systemic loss on both the predicted graph and each MC sample:

$$L_{t,h}(s) := \frac{\sum_e w_e(\hat{G}_{t+h}) - \sum_e w_e(\hat{G}_{t+h}^s)}{\sum_e w_e(\hat{G}_{t+h})}, \quad (12)$$

$$\tilde{L}_{t,h}^{(m)}(s) := \ell(\tilde{G}_{t+h}^{(m)}; s). \quad (13)$$

where $\hat{G}_{t+h}^s$ denotes the network after applying shock s . The scenario engine also tracks the number of *distressed* nodes (those losing $> 50\%$ of total edge weight) and a propagation depth estimate $\lfloor \log_2(\text{affected nodes} + 1) \rfloor + 1$.

CVaR aggregation: To capture tail risk, we compute an empirical CVaR from the MC sample losses. Let $\lambda \in [0, 1]$ (default 0.5) control the tail fraction. We sort the M sample losses in descending order and average the worst $\lceil \lambda M \rceil$:

$$\text{CVaR}_\lambda(s, h) := \frac{1}{\lceil \lambda M \rceil} \sum_{i=1}^{\lceil \lambda M \rceil} \tilde{L}_{t,h}^{(i)}(s), \quad (14)$$

where $\tilde{L}^{(1)} \geq \tilde{L}^{(2)} \geq \dots$ are order statistics. The scenario summary $\mathcal{R}_t$ ranks scenarios by $L_{t,h}(s)$ and retains the worst-case scenario across all horizons.

4.5 Task II: Decision Agent

Action space: The decision module maps alerts and scenario results to recommendation tickets using a conserva-

Action	Severity	Weight w_u	Safe-mode OK?	Min. alerts
Monitor	Low	0.25	Yes	1
Investigate	Medium	0.50	Yes	1
Recommend -Reduce	High	0.75	No	2
Contingency	Critical	1.00	No	3

TABLE 3
Playbook action space $\mathcal{U}$ with severity weights and feasibility constraints.

tive playbook with four action types, ordered by escalation severity:

Feasibility constraints: An intervention-type action u (those requiring safe-mode off) is feasible only when both the data-health gate and the confidence gate are satisfied:

$$u \in \mathcal{C}_t \iff (\text{SAFE_MODE}_t = 0) \wedge (\bar{C}_t \geq \tau_{\text{conf}}), \quad (15)$$

where $\bar{C}_t = \frac{1}{|\mathcal{A}_t|} \sum_{a \in \mathcal{A}_t} C_t(a)$ is the mean alert confidence and τ_{conf} (default 0.6) gates interventions. Non-interventional actions (Monitor, Investigate) are always feasible when at least one alert exists.

Ticket scoring: Each feasible recommendation is scored by a multiplicative heuristic combining playbook severity, alert confidence, and scenario impact:

$$\text{Score}_t(u) := w_u \cdot \bar{C}_t \cdot \text{Imp}_t, \quad (16)$$

where w_u is the severity weight (Table 3), $\bar{C}_t$ is the mean alert confidence, and $\text{Imp}_t := \frac{1}{|\mathcal{S}_0|} \sum_{s \in \mathcal{S}_0} \text{CVaR}_\lambda(s)$ aggregates scenario impact as the mean CVaR across all scenarios. When no scenarios are available, Imp_t defaults to 1.0 (neutral multiplier), ensuring alert-only tickets receive non-zero scores. The agent returns the top- N tickets (default $N = 10$) ranked by $\text{Score}_t(u)$.

Target selection: For intervention tickets, targets are selected by collecting the top- K attributed nodes from all triggering alerts and worst-case scenarios, ranked by frequency-weighted vote.

4.6 End-to-End Procedure

Algorithm 1 summarises the full agent loop.

5 DEXPOSURE-BENCH: AN EVALUATION SUITE FOR FM+LLM DECISION PIPELINES

5.1 Motivation: an evaluation gap

Foundation-model pipelines that pair a domain forecaster with a large language model decision layer are proliferating across finance, supply-chain risk, and infrastructure monitoring. Despite this, the community lacks a standardised suite for evaluating the full *pipeline*—most existing benchmarks evaluate a single stage in isolation. Generic LLM-agent benchmarks (SWE-BENCH [51], AGENTBENCH [52], HELM [50]) cover open-ended coding or tool use but not domain-grounded supervisory decisions. Temporal-graph benchmarks (TGB [23], OGB [48]) measure structural prediction quality but stop short of any decision metric. Domain-specific evaluations of DeFi risk forecasting exist as one-off experiments accompanying single papers, none of them packaged as a reusable harness.

Algorithm 1 DeXposure-Claw: one epoch

Require: State (G_t, X_t) ; horizons $\mathcal{H}$; scenarios $\mathcal{S}_0$; thresholds $(\tau_{\text{data}}, \tau_{\text{conf}}, z, \pi_{\text{min}})$; MC count M
Ensure: Alerts $\mathcal{A}_t$; scenario summary $\mathcal{R}_t$; tickets $\mathcal{D}_t$

- 1: $DH_t \leftarrow \text{DataHealth}(G_t, X_t)$; $\text{SAFE_MODE}_t \leftarrow \mathbb{I}\{DH_t < \tau_{\text{data}}\}$
- 2: **for all** $h \in \mathcal{H}$ **do**
- 3: $\mathcal{P}_{t,h} \leftarrow \text{DEXPOSURE-FM.Forecast}(G_t, X_t, h)$
- 4: $\hat{G}_{t+h} \leftarrow \text{BuildPredGraph}(\mathcal{P}_{t,h}; \pi_{\text{min}})$ $\triangleright$ Eq. (6)
- 5: $\{\tilde{G}_{t+h}^{(m)}\}_{m=1}^M \leftarrow \text{MCSample}(\mathcal{P}_{t,h})$ $\triangleright$ Bernoulli edges + Gaussian weights
- 6: $\mathbf{N}_{t,h} \leftarrow \Phi(\hat{G}_{t+h}, X_t)$; $\tilde{\mathbf{N}}_{t,h}^{(m)} \leftarrow \Phi(\tilde{G}_{t+h}^{(m)}, X_t)$ $\triangleright$ Table 1; $\mathbf{N}_1, \dots, \mathbf{N}_5$
- 7: $(\mu_t^{(j)}, \sigma_t^{(j)}) \leftarrow \text{RollingStats}(\{\mathbf{N}_{t-l,h}\}_{l=1}^L)$ $\triangleright L = 26$ weeks
- 8: $\mathcal{A}_{t,h} \leftarrow \{(h, j) : |N_{t,h}^{(j)} - \mu_t^{(j)}| / \max(\sigma_t^{(j)}, \epsilon) > z\}$ $\triangleright$ Eq. (10)
- 9: Attach $\text{Ev}_t(a)$ and $C_t(a)$ for all $a \in \mathcal{A}_{t,h}$ $\triangleright$ Eq. (11), uses $\tilde{\mathbf{N}}_{t,h}^{(m)}$ for CV
- 10: **for all** $s \in \mathcal{S}_0$ **do**
- 11: $L_{t,h}(s) \leftarrow \text{StressLoss}(\hat{G}_{t+h}, s)$; $\tilde{L}_{t,h}^{(m)}(s) \leftarrow \text{StressLoss}(\tilde{G}_{t+h}^{(m)}, s)$
- 12: $\text{CVaR}_\lambda(s, h) \leftarrow \text{TailMean}(\{\tilde{L}_{t,h}^{(m)}(s)\}_{m=1}^M, \lambda)$ $\triangleright$ Eq. (14)
- 13: **end for**
- 14: **end for**
- 15: $\mathcal{A}_t \leftarrow \bigcup_{h \in \mathcal{H}} \mathcal{A}_{t,h}$
- 16: $\mathcal{R}_t \leftarrow \text{RankScenarios}(\{L_{t,h}(s), \text{CVaR}_\lambda(s, h)\}_{h,s})$
- 17: $\mathcal{D}_t \leftarrow \text{PlaybookDecision}(\mathcal{A}_t, \mathcal{R}_t, DH_t, \text{SAFE_MODE}_t, \tau_{\text{conf}})$ $\triangleright$ Table 3
- 18: **return** $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$

DEXPOSURE-BENCH fills this gap with a single, reproducible suite that scores a candidate pipeline on six axes spanning forecast quality, anomaly detection, predictive uncertainty, what-if query fidelity, downstream decision quality, and data-quality robustness, on a shared 16,610-protocol DeFi exposure dataset with a *regulator-aligned* ground truth.

5.2 Suite schema

The suite is organised as six benchmarks, each producing an independently reportable scalar (or vector of scalars). Pipelines may report against any subset; the full leaderboard requires all six. Table 4 summarises each axis.

The six axes were chosen so that they are: (i) *non-overlapping* in what they measure—a system can excel at b1 while failing b5; (ii) *decomposable along the four-layer framework* (§4), so that ablations on the predictor layer (b1, b3, b6), monitor layer (b2), scenario layer (b4), and decision layer (b5) can be reported independently; and (iii) *individually publishable*, so that downstream work may extend a single axis without re-running the whole suite.

5.3 Dataset and split

The benchmark draws on the DEXPOSURE weekly exposure-graph dataset [4], a directed weighted graph $G_t = (V_t, E_t, w_t)$ at each weekly timestep t , with V_t a subset of

TABLE 4
DEXPOSURE-BENCH schema. Each row is an independently reportable benchmark; the full leaderboard requires all six.

ID	Capability tested	Primary metrics	Applicable methods
b1_forecast	Temporal graph prediction quality	PageRank MAE, HHI MAE, Gini MAE, trend consistency, rank correlation; per horizon $h \in \{1, 4, 8, 12\}$	Any predictor
b2_warning	Streaming anomaly detection	Precision, recall, F1, lead time, stability; per event $\times$ alert budget	Any monitor + predictor
b3_calibration	Predictive uncertainty quality	PI coverage @ 0.90 target, PI width, ECE, CRPS	Predictors with intervals
b4_stress	What-if scenario fidelity	Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@ k ; per scenario $S_1 \dots S_5$	Predictor + scenario engine
b5_decision	Supervisory ticket quality	Precision, recall@ k , F1, target stability, judge score, FIR	Predictor + decision policy
b6_robustness	Data quality sensitivity	Per-benchmark metrics under 5 degradation regimes; relative degradation	Any predictor

TABLE 5
DEXPOSURE-BENCH canonical split. All reported numbers in this paper come from the test split.

Split	Period	Weeks	Usage
Train	2020-03–2024-06	~220	Predictor training
Validation	2024-07–2024-12	~26	Conformal calibration, threshold tuning
Test	2025-01–2025-08	~33	Frozen leaderboard

the 16,610-protocol universe observed over 2020-03–2025-08 (283 weeks). Node features carry log-TVL, number of token types, maximum token share, Shannon entropy of the token distribution, and a 15-class category one-hot. Edge weights are log-scaled USD inter-protocol exposures. We adopt the strict no-leakage temporal split of Table 5.

5.4 Ground truth: regulator-aligned absolute loss

Prior systemic-risk evaluations rank protocols by *fractional* weight change, which produces a ground-truth pool dominated by small protocols whose deterioration carries low systemic significance. This choice silently biases every downstream metric and is not what supervisory authorities monitor in practice.

DEXPOSURE-BENCH therefore defines its ground truth in terms of *absolute* exposure loss. Let $w_t(v) = \sum_{e \ni v} w_t(e)$ be the total weight incident on protocol v at time t . For horizon h and lookahead window W we define the *stressed set* $\mathcal{S}_t^h$ as the top π fraction of active protocols ranked by absolute loss:

$$\begin{aligned} \Delta_t^h(v) &= w_t(v) - w_{t+h}(v), \\ \mathcal{S}_t^h &= \text{top}_\pi \{v : w_t(v) > 0, \Delta_t^h(v) > 0\}. \end{aligned} \quad (17)$$

We fix $\pi = 0.05$ throughout (yielding an average pool of $|\mathcal{S}_t^h| \approx 283$ protocols per week, stable across the test split) and the lookahead window $W = 4$ weeks.

Equation 17 is the single ground-truth definition used by b2, b5, and the LLM-decision evaluation pipeline. Aligning the metric with absolute economic magnitude—rather than fractional change—restores discrimination between methods that target central versus peripheral nodes and matches how a supervisor would prioritise.

5.5 Methods supported out of the box

Eight reference methods are bundled with the harness, sufficient to populate a complete leaderboard from a fresh

checkout. Each method exposes the same `predict_graph` or `decide_actions` interface, so adding a ninth method is a single-file change.

The eight listed in Table 6 together expose all four cuts of the framework ablation: (i) adding the predictor layer (`m1`→`m5`), (ii) swapping the decision layer (`m5`→`m6`), (iii) adding the safety gate (`m6`→`m7`), and (iv) removing the predictor from the LLM stack (`m2`→`m6`).

5.6 Reporting protocol

To enter the leaderboard, a submission produces one JSON file per {benchmark, method} pair following the harness’s published schema, and a single `results.json` aggregating the suite. Required reports for the headline table are b1, b3, b5, and b6; b2 and b4 are required only for systems that implement a monitor or scenario layer respectively. All metrics are computed on the frozen test split of Table 5; training and hyperparameter tuning may use train + validation freely. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files so that any reported number can be reproduced bit-exactly.

5.7 Positioning in the benchmark landscape

Table 7 compares DEXPOSURE-BENCH against the closest existing evaluation suites. No prior suite covers a domain-grounded decision metric on top of a forecasting backbone, which is the niche DEXPOSURE-BENCH addresses.

5.8 Reproducibility and extensibility

The harness, dataset, and reference implementations are released under an open licence at the project repository. A complete leaderboard run, including all eight reference methods and the LLM evaluation pipeline, completes in under four GPU-hours on a single RTX 4090 plus approximately ten USD of LLM API spend, making the benchmark accessible to academic groups without industrial compute. Extending the suite—adding a method, an additional ground-truth formulation, or a new benchmark axis—is supported by single-file plug-in points documented alongside the harness.

TABLE 6
Reference methods bundled with DEXPOSURE-BENCH. Heuristic identifiers use the h namespace; learned methods use m .

ID	Predictor	Policy	Role
$h1_weighted_degree$	—	heuristic alert	Streaming anomaly baseline
$m1_persistence_rules$	$G_{t+h} = G_t$	rule engine	Decision baseline
$m2_snapshot_llm$	current snap	LLM	LLM-only ablation
$m3_evolvegcnn$	EvolveGCN	—	Learned-GNN baseline
$m4_fm_only$	FM	—	FM-only ablation
$m5_fm_rules$	FM	rule engine	FM + rules
$m6_fm_llm$	FM	LLM	Full FM + LLM stack
$m7_fm_llm_gated$	FM	LLM + safety gate	Safety-gated FM + LLM

TABLE 7
DEXPOSURE-BENCH versus closest existing suites. A check mark indicates a covered axis; — indicates the axis is out of scope.

Suite	Pred.	Anom.	Calib.	Scen.	Decision
OGB [48]	✓	—	—	—	—
TGB [23]	✓	—	—	—	—
HELM [50]	—	—	—	—	✓
SWE-BENCH [51]	—	—	—	—	✓
AGENTBENCH [52]	—	—	—	—	✓
DEXPOSURE-BENCH (this work)	✓	✓	✓	✓	✓

6 EXPERIMENTS

We evaluate DEXPOSURE-CLAW on the DEXPOSURE-BENCH suite (§5), grouping the six benchmarks under two natural axes: (i) does the forecast-measure pipeline produce useful, calibrated, and robust risk signals (Task I, covering b1, b2, b3, b6), and (ii) do the resulting decision tickets improve the recall-explanation operating point while exposing their intervention cost (Task II, covering b4, b5)? The result narrative follows the structure of the benchmark itself: Fig. 3 establishes the six-axis evaluation stage, Table 8 and Fig. 4 audit where the FM helps and where persistence remains stronger, Table 9 and Fig. 5 isolate the layer-wise contribution and cost of FM, LLM, and safety gate, and the timeline/ablation tables close with reliability evidence. Unless explicitly marked as a historical event study, results are reported on the frozen 2025 test split of Table 5; the ground truth, dataset split, and reference methods used throughout are defined in §5.

6.1 Task I: Network Risk Monitoring

6.1.1 Implementation Details

Data and temporal granularity: We use the DEXPOSURE dataset [4], comprising weekly directed weighted exposure-graph snapshots sourced from the DeFiLlama API. The dataset spans 2020-03 to 2025-08 (283 weeks), with 50–500 protocol nodes and 100–2,000 exposure edges per snapshot, totalling over 43.7M data entries. Node features include log-TVL, number of token types, maximum token share, Shannon entropy of token distribution, and protocol category (one-hot, 15 classes). Edge weights are log-scaled USD exposure values. We evaluate multi-horizon forecasting on $\mathcal{H} = \{1, 4, 8, 12\}$ weeks.

Temporal split: We enforce a strict no-leakage temporal split:

Split	Period	Weeks	Usage
Train	2020-03 – 2024-06	~222	FM backbone training
Val	2024-07 – 2024-12	~26	Hyperparameter + threshold tuning
Test	2025-01 – 2025-08	~33	Final evaluation

Stress event windows: For early-warning evaluation ($b2_warning$), we additionally run a historical event study around three major DeFi crises that occur before the 2025 test split. This diagnostic evaluates whether the monitor turns before known shocks; it is reported separately from the predictor-specific 2025 leaderboard:

Event	Date	Pre-shock (4w)	Event (3w)
Terra/Luna	2022-05-09	03-28 – 05-02	05-02 – 05-23
FTX	2022-11-07	09-26 – 10-31	10-31 – 11-21
SVB/USDC	2023-03-10	02-06 – 03-06	03-06 – 03-27

Forecast interface and predicted-graph construction: For each epoch t and horizon $h \in \mathcal{H}$, we query the backbone model to obtain a predictive distribution $\mathcal{P}_{t,h}$. We construct an expected-weight predicted graph $\hat{G}_{t+h}$ by thresholding edge existence probabilities at $\pi_{\min}$ and setting $\hat{w}_{pq} := \Pr(e_{pq} = 1) \cdot \mathbb{E}[w_{pq} \mid e_{pq} = 1]$ (Eq. (6)). We set $\pi_{\min} = 0.2$ on the validation split. For uncertainty quantification, we draw $M = 50$ Monte Carlo samples via Bernoulli edge sampling and Gaussian weight perturbation.

Monitoring metrics and alert triggers: At each horizon, we compute five monitoring functionals $N_{t,h} = \Phi(\hat{G}_{t+h}, X_t)$ (Table 1): systemic importance (N1), HHI concentration (N2), network density (N3), PageRank Gini (N4), and degree Gini (N5). Baseline bands $(\mu_t^{(j)}, \sigma_t^{(j)})$ are computed via a rolling window of $L = 26$ weeks. An alert fires when a metric exceeds $z = 2$ standard deviations from baseline (Eq. (10)). Alert confidence follows Eq. (11).

Stress scenarios: We evaluate five standardised scenarios S_0 (Table 2): single protocol failure (S1, 100% shock on top node), bridge cluster failure (S2, 100% on bridges), stablecoin de-peg (S3, 50% on stablecoins), sector-wide lending shock (S4, 30%), and correlated stress (S5, 20% on top-10 nodes). CVaR is computed as the mean of the worst $\lambda = 0.5$ fraction of MC sample losses (Eq. (14)).

Attribution: For each triggered alert, we rank contributing edges by marginal contribution: $\text{Contrib}(e) := \Phi^{(j)}(\hat{G}_{t+h}) - \Phi^{(j)}(\hat{G}_{t+h} \setminus e)$, retaining the top $K = 10$.

Evaluation protocol: Within the six-axis framework of Fig. 3, Task I evaluates the forecast-measure pipeline along three axes:

- 1) *Risk-metric forecasting ($b1_forecast$):* predicted vs. realised metrics using MAE and Spearman rank correlation, for each $h \in \{1, 4, 8, 12\}$.

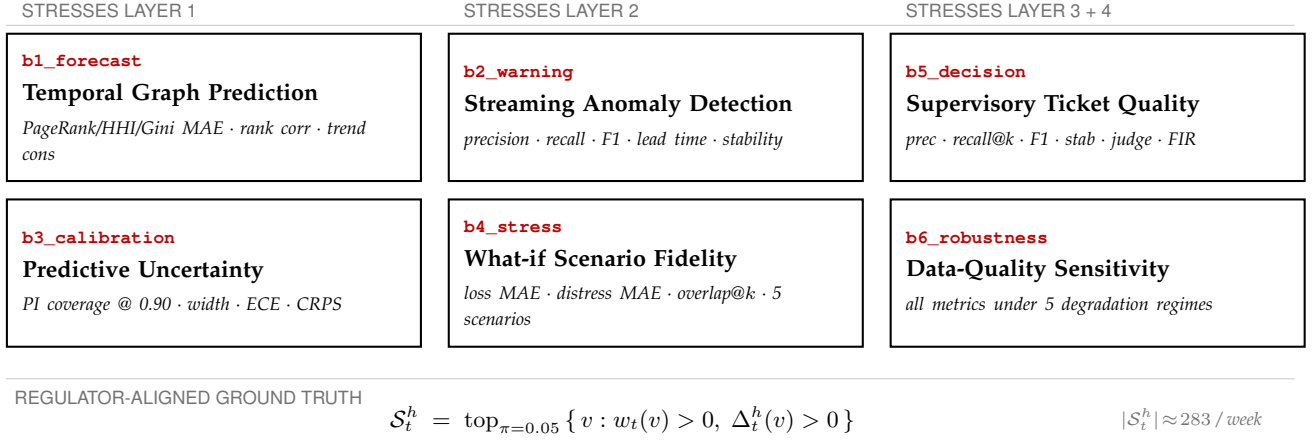


Fig. 3. DEXPOSURE-BENCH: a six-axis evaluation suite. The six benchmarks are grouped by which framework layer they stress; the bottom strip carries the regulator-aligned ground truth that anchors all decision-quality metrics.

- 2) *Early-warning quality (b2_warning)*: lead time to stress events, precision/recall under alert budgets $\{5, 10, 20\}$, alert stability (flip rate), and F1-warning.
- 3) *Uncertainty calibration (b3_calibration)*: expected calibration error (ECE), prediction interval coverage (target 90%), PI width, and CRPS. We apply split conformal prediction on the validation set to ensure asymptotically valid coverage.

Reproducibility: All hyperparameters ($\pi_{\min}$, L , z , K , M , τ_{data}) are selected on the validation split and fixed for test evaluation. Random seeds are fixed for MC sampling and LLM consistency checks. The pipeline is fail-closed on predictors: an unavailable FM or EvolveGCN checkpoint aborts the corresponding run rather than silently falling back to persistence. For *b2_warning* we report the shared weighted-degree heuristic (*h1_weighted_degree*), since *b2*'s monitoring layer sits upstream of the predictor choice and is therefore not a method-specific comparison.

6.1.2 Empirical Results

Table 8 reports the Task I evidence ledger used in the main text; the full all-horizon *b1_forecast* matrix is provided as Table 12 in Appendix A. The point of the table is not to claim that the FM wins every forecasting metric. Instead, it separates three types of evidence: 2025 predictor-specific evidence (*b1*, *b3*, *b6*), a historical monitor event study (*b2*), and the strong persistence baseline that any deployable monitor must acknowledge.

Persistence is a deceptively strong point-metric baseline: On PageRank MAE, Gini MAE, and rank correlation, persistence matches or marginally beats the FM at every horizon (rank correlation 0.570 vs 0.558 at $h=4$; PR MAE 3.4 vs 4.5 in 10^{-5} units). Weekly DeFi exposure structure is heavily auto-correlated, so any predictor that returns the most recent graph captures the dominant source of variance “for free”—the same phenomenon that makes seasonal-naïve a strong baseline in classical forecasting. A new model must clear this bar before its other properties matter.

FM gains are concentrated in trend and robustness, not in absolute error: The FM produces a non-trivial trend signal (trend consistency 0.525–0.632 across horizons) while

persistence is structurally pinned at 0. Under data degradation (*b6_robustness*, mean relative degradation 0.113 vs 0.148 for persistence), the FM drops 24% less on average. These results support a narrower claim than pointwise superiority: the FM contributes direction, uncertainty, and robustness features that persistence does not provide, while persistence remains the better static point-metric baseline.

Calibration is strong; learned GNN transfer remains difficult: The FM achieves PI coverage 0.913 against a 0.90 target with ECE 0.013 on *b3_calibration*, both within the expected slack of split-conformal calibration. Persistence has no native uncertainty estimate and is omitted from this row. EvolveGCN is generally weaker than persistence and FM variants on the downstream metrics in the appendices, which we attribute to data efficiency: 222 weekly training snapshots fall below the regime where it learns a useful structural prior, while the FM benefits from large-scale pre-training transfer.

6.2 Task II: Decision-Making

6.2.1 Implementation Details

Decision timeline and inputs: Task II operates at the same decision epochs as Task I. At each epoch, the decision agent receives: (i) the current state (G_t, X_t) , (ii) multi-horizon forecasts $\{\hat{G}_{t+h}\}_{h \in \mathcal{H}}$ and optional MC samples, (iii) monitoring alerts $\mathcal{A}_t$ with evidence and confidence scores, and (iv) the scenario summary $\mathcal{R}_t$.

Action space and playbook: We use a four-level conservative playbook (Table 3): *Monitor* (severity weight 0.25, always feasible), *Investigate* (0.50, always feasible), *Recommend-Reduce* (0.75, requires $\text{SAFE_MODE} = 0$ and ≥ 2 alerts), and *Contingency* (1.00, requires $\text{SAFE_MODE} = 0$ and ≥ 3 alerts). Intervention-type actions additionally require mean alert confidence $\bar{C}_t \geq \tau_{\text{conf}} = 0.6$.

Ticket scoring: Tickets are ranked by $\text{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \text{Imp}_t$ (Eq. (16)), where w_u is the playbook severity weight, $\bar{C}_t$ is mean alert confidence, and Imp_t is the mean CVaR across all scenarios (defaulting to 1.0 when no scenarios are available). We output the top- $N = 10$ tickets.

Evidence item	Scope / method	Value	Comparator
PR-MAE@h4 (10^{-5} , lower better)	2025 b1, FM	4.5	Persist 3.4
RankCorr@h4	2025 b1, FM	0.558	Persist 0.570
TrendCons@h4	2025 b1, FM	0.628	Persist 0.000
HHI-MAE@h4, lower better	2025 b1, FM	0.032	Persist 0.032
Precision, mean	Historical b2, shared h1	0.756	N/A
F1-warning, mean	Historical b2, shared h1	0.007	N/A
Lead time, weeks	Historical b2, shared h1	4.667	N/A
ECE	2025 b3, FM intervals	0.013	N/A
PI coverage	2025 b3, FM intervals	0.913	target 0.900
Relative degradation, mean	2025 b6, FM	0.113	Persist 0.148

TABLE 8

Task I evidence ledger. The 2025 rows compare FM and persistence on forecasting, calibration, and robustness; the b2_warning rows are a separate historical event study using the shared h1_weighted_degree monitor and are not attributed to a specific predictor.

Task I: FM Signal Audit

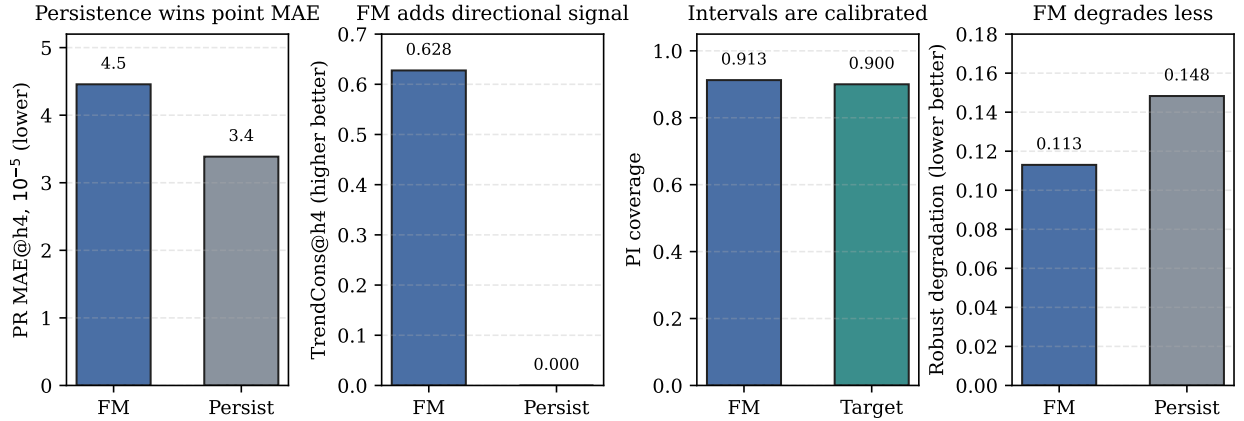


Fig. 4. Task I signal audit. Persistence remains stronger on static point metrics such as PageRank MAE and rank correlation, while the FM contributes directional trend signal, calibrated intervals, and lower mean degradation under corruption.

Target selection: For intervention tickets, targets are selected from the union of top- K attributed nodes across triggering alerts and worst-case scenarios, ranked by frequency-weighted vote.

Evaluation protocol: We evaluate Task II using both offline and scenario-based criteria:

- 1) *Offline (b5_decision):* ticket precision (fraction followed by real stress), target stability (Jaccard between adjacent epochs), audit completeness (evidence bundle coverage), suppression rate (safe-mode activation), and false intervention rate.
- 2) *Scenario-based (b4_stress):* stress-test contagion prediction accuracy—loss MAE, distressed count MAE, propagation depth MAE, and target overlap@ K between predicted and actual networks under scenarios S1–S5.

Reproducibility: All decision hyperparameters (K , N , τ_{data} , τ_{conf} , λ) are tuned on the validation split and fixed for testing. MC components use fixed seeds; LLM consistency is measured through repeated deterministic calls and reported in Table 9.

6.2.2 Empirical Results

Figure 5 is the central layer-wise result: it isolates what the FM, LLM decision layer, and safety gate each add to the decision pipeline, using ticket F1 to combine precision and

recall, LLM-as-judge score to track explanation quality, and false-intervention rate to make the cost visible.

Table 9 then reports the full decision-quality audit across five method configurations. The table should be read as a multi-objective comparison rather than a single leaderboard: m1_persistence_rules is the highest-precision conservative baseline, m5_fm_rules gives the strongest rule-based F1 balance, and the LLM variants trade false-intervention cost for higher recall and explanation quality.

Persistence+Rules wins on single-metric precision: m1_persistence_rules achieves the highest ticket precision (0.720) and, because its false intervention rate is 0, also the highest FIR-penalised cost-adjusted score (0.720). This is consistent with conservative rule-engine behaviour: it raises few tickets and most are correct. It is the right baseline for any system that wants to claim improvement on decision quality.

FM+LLM triples recall but at a real precision cost: m6_fm_llm raises recall from 0.004 (persistence) to 0.012—a $3\times$ improvement on the regulator-aligned recall axis—at the cost of precision (0.570 vs 0.720) and a substantial false intervention rate (FIR 0.448). The safety gate (m7_fm_llm_gated) partially mitigates this and raises the LLM-as-judge score from 2.48 to 2.69/5, but does not fully recover precision. Both LLM variants reach grounding 1.000, so the failure mode is not hallucination—it is over-reaching to

Task II: Recall and Explanation Gains With Explicit Cost

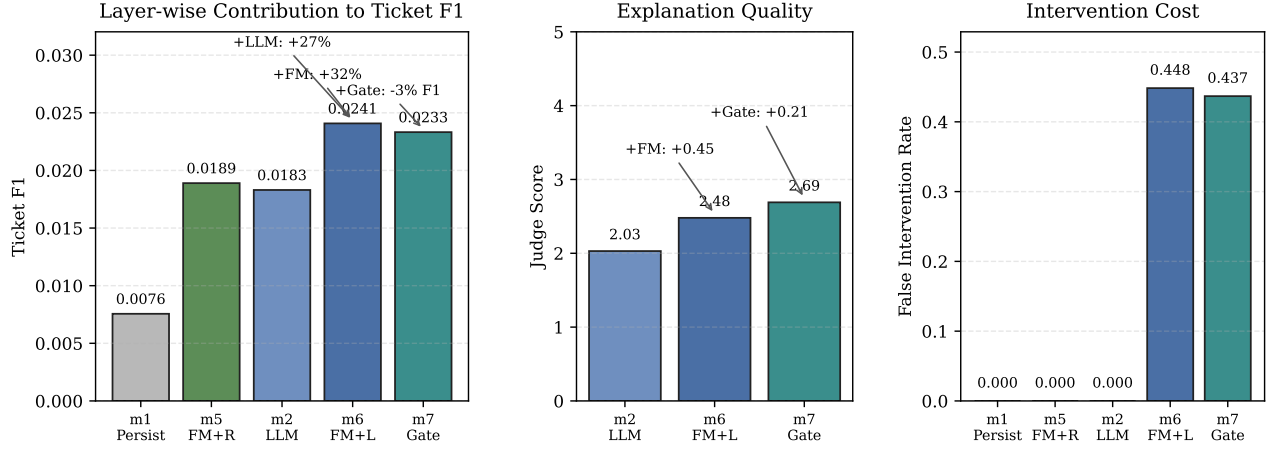


Fig. 5. Task II layer-wise contribution and cost. Adding the FM signal to a snapshot LLM improves ticket F1 and explanation quality; replacing the FM rule engine with the LLM improves recall-oriented F1 further; the same LLM variants incur a substantial false-intervention rate, partially mitigated by the safety gate.

Method	Prec	Recall	F1	Stabil	SevRho	FIR	Ground	Consist	Explain	CostAdj
m1_persistence_rules Persist+Rules	0.720	0.004	0.0076	0.514	N/A	0.000	N/A	N/A	N/A	0.720
m5_fm_rules FM+Rules	0.600	0.010	0.0189	0.257	N/A	0.000	N/A	N/A	N/A	0.600
m2_snapshot_llm Pure LLM	0.575	0.009	0.0183	0.532	N/A	0.000	1.000	0.916	2.0/5	0.575
m6_fm_llm FM+LLM	0.570	0.012	0.0241	0.488	-0.001	0.448	1.000	0.779	2.5/5	0.314
m7_fm_llm_gated FM+LLM+RulesGate	0.580	0.012	0.0233	0.435	-0.003	0.437	1.000	0.855	2.7/5	0.327

TABLE 9

Task II decision-quality comparison. F1 combines Prec and Recall against the regulator-aligned absolute-loss ground truth; absolute values are small because each week's stressed pool S_t^h contains ~ 283 protocols while the agent emits 0.7–2 tickets per week, so recall is budget-bounded by construction. Higher Prec, Recall, F1, Stabil, Ground, Consist, Explain, and CostAdj are better; lower FIR is better.

Note: m3_evolvegc and m4_fm_only are predictor-only baselines and do not emit supervisory tickets, so they are not applicable to b5_decision.

high-severity actions when monitor evidence is weak.

Layer-wise decomposition: Holding the decision policy fixed at LLM and adding the FM signal lifts ticket F1 by 32% (0.0183 $\rightarrow$ 0.0241); holding the predictor fixed at FM and swapping rules for LLM lifts F1 by another 27% (0.0189 $\rightarrow$ 0.0241). The two layers therefore make additive, non-substitutable contributions to F1. Adding the safety gate trades 3% F1 for +0.21 judge score, which is the operating point a deployment operator should select when explanation quality is the binding constraint.

Early-warning timeline analysis: To determine whether the weighted-degree monitoring alerts (h1_weighted_degree) constitute genuine *early warnings* or merely post-hoc damage reports, we conduct a per-week historical alert timeline analysis around the Terra/Luna (2022-05-09), FTX (2022-11-07), and SVB/USDC (2023-03-10) events. Table 10 summarises the results.

The data supports pre-event lead time for this monitor: median lead times are 4–5 weeks across all three crises, with SVB/USDC reaching precision 1.000 at every alert budget. This should be read as historical monitor evidence, not as an FM-specific test-split result. F1 is small in absolute terms because the regulator-aligned ground-truth pool S_t^h contains ~ 283 protocols per week, so a budget of 5–20 alerts covers only a thin slice of the recall surface.

Stress-test fidelity (b4_stress): Per-scenario stress-test outputs are reported in Table 13 (Appendix B). As with

TABLE 10

b2_warning historical early-warning event study for three major DeFi crises (Terra/Luna 2022-05, FTX 2022-11, SVB-USDC 2023-03), evaluated with the shared h1_weighted_degree heuristic across three alert budgets. Lead time reports the median number of weeks alerts precede the event; precision is computed against the 4-week pre-shock window. This table evaluates the monitor/timeline protocol rather than the 2025 predictor-specific leaderboard.

Event	Budget	Lead (wk)	Prec.	Recall	F1	Stability
Terra/Luna	5	5.0	0.600	0.0018	0.0036	0.440
	10	5.0	0.700	0.0033	0.0065	0.620
	20	5.0	0.650	0.0080	0.0157	0.740
FTX	5	5.0	0.600	0.0013	0.0027	0.640
	10	5.0	0.700	0.0024	0.0048	0.720
	20	5.0	0.550	0.0048	0.0095	0.720
SVB / USDC	5	4.0	1.000	0.0018	0.0036	0.550
	10	4.0	1.000	0.0028	0.0056	0.750
	20	4.0	1.000	0.0056	0.0112	0.813

b1_forecast, persistence is a strong baseline because each scenario applies a structural shock to the current graph and the realised loss vector therefore correlates strongly with the persistence prediction. The FM variants roughly match persistence on Loss MAE and Overlap@10 and sometimes marginally improve loss MAE, but the differences are small; this table is best read as evidence that the scenario engine remains competitive with persistence rather than as

TABLE 11

Component-level ablation of DEXPOSURE-CLAW on `b5_decision`. Panel A shows clean 2025 test-split ablations against the `m5_fm_rules` baseline; Panels B–C report targeted stress runs for mechanisms that are dormant on clean data. A2 (confidence gating) and A3 (scenario engine) are load-bearing in the clean setting, while A1 (data-health gating) and A6 (multi-horizon monitoring) are safety-reserve mechanisms whose effects appear under data degradation and crisis-window dynamics, respectively.

Panel	Ablation	Configuration	Ticket Prec.	FIR	Notes
A: clean test	—	full <code>m5_fm_rules</code>	0.600	0.000	baseline
	A1	disable data-health gate ($\tau_{\text{data}}=0$)	0.600	0.000	no effect (clean data)
	A2	disable confidence gate ($\tau_{\text{conf}}=0$)	0.600	0.429	FIR shoots up
	A3	skip scenario engine	0.000	0.000	target extraction collapses
	A6	single-horizon forecasting ($h=4$)	0.600	0.000	no effect (clean data)
B: degraded data	A1 off	80–98% feature/edge mask, $\tau_{\text{conf}}=0$	—	0.27–0.60	false interventions emerge
	A1 on	same degradation, safe-mode triggered	—	0.000	intervention suppression
C: crisis period	A6: single	horizons = {4}, crisis windows	0.533–0.624	0.000	alerts: 19–23
	A6: multi	horizons = {1, 4, 8, 12}, same windows	0.533–0.624	0.000	alerts: 73–98 ($\approx 4\times$)

a sweeping FM win. EvolveGCN is generally weaker on target overlap and distressed-count fidelity, mirroring its weakness on `b1_forecast`.

6.3 Ablation Study

We separate clean-data load-bearing components from stress-reserve mechanisms through ablation experiments under standard and targeted stress conditions (Table 11).

Standard conditions (Panel A): Under clean 2025 test data, the confidence gate (A2) is load-bearing: removing it increases the false intervention rate from 0.000 to 0.429. The scenario engine (A3) is also load-bearing for target extraction: skipping it reduces ticket precision from 0.600 to 0.000. Data-health gating (A1) and multi-horizon diversity (A6) show no measurable impact in this clean setting because data quality is high and $h = 4$ dominates alert generation.

Stress conditions (Panels B–C): To test whether A1 and A6 are unnecessary or merely dormant on clean data, we evaluate them under targeted stress conditions. For A1, we apply severe feature/edge degradation regimes (80–98% masking) and isolate the data-health gate from the confidence gate by setting $\tau_{\text{conf}} = 0$. Under these conditions, disabling the data-health gate increases the false intervention rate to 0.27–0.60 (Panel B), while triggered safe-mode suppresses interventions back to 0.000, showing that A1 functions as a *safety reserve under data degradation*. For A6, we compare single-horizon ($h = 4$) against multi-horizon ($h \in \{1, 4, 8, 12\}$) monitoring across the Terra/Luna, FTX, and SVB/USDC crisis windows: multi-horizon generates roughly $4\times$ more alerts (73–98 vs. 19–23) while maintaining the same precision range (0.533–0.624) and zero false interventions (Panel C). Thus A6 broadens alert coverage in crisis windows, but Table 11 does not show a clean-data precision gain. The ablation evidence therefore supports a two-part interpretation: A2 and A3 carry the clean deployment path, while A1 and A6 are reserve mechanisms for degraded-data and crisis-window operation.

7 LIMITATIONS

We disclose three limitations that bound the scope of our claims.

Single dataset: All results are reported on the DEXPOSURE weekly exposure-graph dataset. While it covers 4,300 protocols, 24,300 tokens, 43.7M exposure entries, and 283 weekly snapshots spanning the major DeFi crisis episodes of 2020–2025, generalisation to other on-chain risk surfaces—NFT lending, perpetual derivatives, or cross-chain bridge networks—is untested and will require domain-specific re-calibration of the conformal split and the stressed-set percentile π .

Weekly temporal resolution: The forecast and decision pipeline operates at weekly granularity. Many DeFi shocks unfold within hours—the Terra/Luna collapse erased \$40B in under 48 hours—and a production deployment would require sub-hourly snapshots and re-evaluation of conformal calibration on the resulting denser time series. The historical `b2_warning` lead times of 4–5 weeks should therefore be read as “how early the weekly monitor turns”, not as end-to-end alert latency.

Offline decision evaluation: `b5_decision` scores tickets against a regulator-aligned absolute-loss ground truth, and the explanation-quality metric is an LLM-as-judge proxy with Claude Opus 4.7 as judge. Neither captures the downstream effect of supervisory action—whether the recommended risk reduction would actually have averted contagion. A controlled live evaluation with a regulator in the loop, which the bench approximates but cannot replace, remains the primary direction we plan to extend the benchmark in.

8 CONCLUSION

We presented DEXPOSURE-CLAW, DEXPOSURE-BENCH, and an empirical characterisation of FM + LLM risk-decision pipelines on temporal DeFi exposure graphs. DEXPOSURE-CLAW turns predicted exposure graphs into auditable supervisory tickets through four layers: FM prediction, deterministic monitor and scenario engines, LLM decision, and a data-health/confidence safety gate. DEXPOSURE-BENCH, released alongside, provides a six-axis evaluation suite with eight reference method implementations and a regulator-aligned absolute-loss ground truth on the 4,300-protocol DEXPOSURE corpus.

The core empirical finding is that the full agentic stack improves the recall–explanation operating point while mak-

ing its cost explicit. On the 2025 decision split, DEXPOSURE-CLAW raises end-to-end ticket F1 from 0.0076 (persistence + rules) to 0.0233 (+208%), and LLM-as-judge explanation quality from 2.03 to 2.69/5 (+33%). The forecasting layer is not a point-metric sweep over persistence: persistence remains stronger on PageRank MAE and static rank correlation, while the FM contributes trend signal, calibrated intervals (coverage 0.913, ECE 0.013), and lower mean degradation under corruption (0.113 vs. 0.148). A separate historical event study shows that the shared weighted-degree monitor turns 4–5 weeks before Terra/Luna, FTX, and SVB-USDC, but this is monitor/timeline evidence rather than a predictor-specific 2025 leaderboard result. The decision gain is a Pareto trade-off rather than uniform dominance: FM+LLM variants raise recall and explanation at a non-trivial false-intervention cost ($FIR \approx 0.44$) compared with rule-based baselines ($FIR = 0$). Layer-wise attribution—+32% F1 from adding FM to a snapshot LLM, +27% from swapping rules for LLM under a fixed FM—shows that the predictor and decision layers are additive on recall-oriented F1; the safety gate then trades 3% F1 for +0.21 judge score and +10% consistency, defining the auditability operating point.

Deployability characterisation: Beyond aggregate metrics, DEXPOSURE-BENCH records the deployment properties that determine whether the system can be audited and operated. The FM+monitor+scenario pipeline (`m5_fm_rules`) processes one weekly snapshot in 67.6 s on a single RTX 4090; the LLM decision layer adds an external API call whose latency is reported but not treated as a GPU benchmark. The ablations separate clean-data necessities from reserve mechanisms: removing the scenario engine collapses ticket precision to 0, and disabling the confidence gate lifts FIR from 0 to 0.429 on clean 2025 data. Under 80–98% feature/edge masking, the data-health gate becomes the fail-safe: without it, FIR rises to 0.29–0.60; with strict safe-mode suppression, FIR returns to 0. Multi-horizon monitoring is best read as a crisis-window coverage reserve, expanding alert counts by roughly $4\times$ during Terra/Luna, FTX, and SVB-USDC windows while preserving the same precision range. Every ticket carries the FM prediction, monitor scores, scenario projections, and, when applicable, LLM rationale that produced it; we release these audit logs alongside the harness.

Beyond the limitations in §7, two directions are most pressing. First, supervised use of the bench as a reward signal to fine-tune a regulator-specialised small model from the LLM rationales we release, closing the loop between offline evaluation and on-policy deployment. Second, extending the harness to additional on-chain risk surfaces—NFT lending, perpetual derivatives, cross-chain bridges—so the FM+LLM+gate template can be benchmarked beyond DeFi exposure, alongside a controlled regulator-in-the-loop study at sub-weekly resolution. We release the harness, dataset, eight reference implementations, and the ticket-level audit logs to invite the community to do both.

REFERENCES

- [1] D. Yao, S. Jagannath, B. Aroso, V. Krishnan, and D. Zhao, “Auditable LLM arbiter for DeFi security: A hybrid graph-of-thoughts approach to intent-transaction alignment,” in *NDSS 2026 LAST-X Workshop*, 2026.
- [2] Q. Mao, Y. Zhang, J. Chen, W. Zhou, and J. Yan, “Know your intent: An autonomous multi-perspective LLM agent framework for DeFi user transaction intent mining,” 2025. [Online]. Available: <https://arxiv.org/abs/2511.15456>
- [3] X. Li, X. Shan, W. Yin, and H. Wang, “Heterogeneous graph pre-training based model for secure and efficient prediction of default risk propagation among bond issuers,” 2025. [Online]. Available: <https://arxiv.org/abs/2501.03268>
- [4] W. Wu, K. Qian, A. Lui, C. Jack, Y. Wu, P. McBurney, F. He, and B. Zhang, “Dexposure: A dataset and benchmarks for inter-protocol credit exposure in decentralized financial networks,” 2025. [Online]. Available: <https://arxiv.org/abs/2511.22314>
- [5] L. Eisenberg and T. H. Noe, “Systemic risk in financial systems,” *Management Science*, vol. 47, no. 2, pp. 236–249, Feb. 2001. [Online]. Available: <https://pubsonline.informs.org/doi/10.1287/mnsc.47.2.236.9835>
- [6] D. Acemoglu, A. Ozdaglar, and A. Tahbaz-Salehi, “Systemic risk and stability in financial networks,” *American Economic Review*, vol. 105, no. 2, p. 564–608, February 2015. [Online]. Available: <https://www.aeaweb.org/articles?id=10.1257/aer.20130456>
- [7] P. Glasserman and H. P. Young, “Contagion in financial networks,” *Journal of Economic Literature*, vol. 54, no. 3, p. 779–831, September 2016. [Online]. Available: <https://www.aeaweb.org/articles?id=10.1257/jel.20151228>
- [8] S. Battiston, M. Puliga, R. Kaushik, P. Tasca, and G. Caldarelli, “Debtrank: Too central to fail? financial networks, the fed and systemic risk,” *Scientific Reports*, vol. 2, p. 541, Aug. 2012. [Online]. Available: <https://www.nature.com/articles/srep00541>
- [9] F. Schär, “Decentralized finance: On blockchain- and smart contract-based financial markets,” *Federal Reserve Bank of St. Louis Review*, vol. 103, no. 2, pp. 153–174, 2021. [Online]. Available: <https://www.stlouisfed.org/publications/review/2021/02/05/decentralized-finance-on-blockchain-and-smart-contract-based-financial-markets>
- [10] K. Gogol, C. Killer, M. Schlosser, T. Bocek, B. Stiller, and C. Tessone, “Sok: Decentralized finance (defi) – fundamentals, taxonomy and risks,” 2024. [Online]. Available: <https://arxiv.org/abs/2404.11281>
- [11] European Systemic Risk Board, “Crypto-assets and decentralised finance: Report on stablecoins, crypto-investment products and multi-function groups,” European Systemic Risk Board, Tech. Rep., Oct. 2025, accessed 27 December 2025. [Online]. Available: https://www.esrb.europa.eu/pub/pdf/reports/esrb-report202510_cryptoassets.en.pdf
- [12] J. Bertomeu, X. Martin, and I. Sall, “Measuring defi risk,” *Finance Research Letters*, vol. 63, p. 105321, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1544612324003519>
- [13] S. Aufiero, S. Bartolucci, F. Caccioli, and P. Vivo, “Mapping microscopic and systemic risks in tradfi and defi: a literature review,” 2025. [Online]. Available: <https://arxiv.org/abs/2508.12007>
- [14] S. Zhang, Z. Wang, J. Zheng, and J. Cartlidge, “Systemic risk in DeFi: A network-based fragility analysis of TVL dynamics,” *arXiv preprint arXiv:2601.08540*, 2026. [Online]. Available: <https://arxiv.org/abs/2601.08540>
- [15] V. Balmaseda, M. Coronado, and G. de Cadenas-Santiago, “Predicting systemic risk in financial systems using deep graph learning,” *Intelligent Systems with Applications*, vol. 19, p. 200240, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2667305323000650>
- [16] L. Gonon, T. Meyer-Brandis, and N. Weber, “Computing systemic risk measures with graph neural networks,” 2025. [Online]. Available: <https://arxiv.org/abs/2410.07222>
- [17] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” 2017. [Online]. Available: <https://arxiv.org/abs/1609.02907>
- [18] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph attention networks,” 2018. [Online]. Available: <https://arxiv.org/abs/1710.10903>
- [19] A. Sankar, Y. Wu, L. Gou, W. Zhang, and H. Yang, “Dysat: Deep neural representation learning on dynamic graphs via self-attention networks,” in *Proceedings of the 13th International Conference on Web Search and Data Mining*, 2020, pp. 519–527.
- [20] D. Xu, C. Ruan, E. Körpeoğlu, S. Kumar, and K. Achan, “Inductive representation learning on temporal graphs,” in *International*

- Conference on Learning Representations (ICLR)*, 2020. [Online]. Available: <https://openreview.net/forum?id=rJeW1yHYwH>
- [21] A. Pareja, G. Domeniconi, J. Chen, T. Ma, T. Suzumura, H. Kanezashi, T. Kaler, T. Schardl, and C. Leiserson, "EvolveGCN: Evolving graph convolutional networks for dynamic graphs," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 04, 2020, pp. 5363–5370.
 - [22] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein, "Temporal graph networks for deep learning on dynamic graphs," 2020. [Online]. Available: <https://arxiv.org/abs/2006.10637>
 - [23] S. Huang, F. Poursafaei, J. Danovitch, M. Fey, W. Hu, E. Rossi, J. Leskovec, M. Bronstein, G. Rabusseau, and R. Rabbany, "Temporal graph benchmark for machine learning on temporal graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
 - [24] R. Bommasani, D. A. Hudson, E. Adeli, R. Altman, S. Arora, S. von Arx, M. S. Bernstein, J. Bohg, A. Bosselut, E. Brunskill, E. Brynjolfsson, S. Buch, D. Card, R. Castellon, N. Chatterji, A. Chen, K. Creel, J. Q. Davis, D. Demszky, C. Donahue, M. Doumbouya, E. Durmus, S. Ermon, J. Etchemendy, K. Ethayarajh, L. Fei-Fei, C. Finn, T. Gale, L. Gillespie, K. Goel, N. Goodman, S. Grossman, N. Guha, T. Hashimoto, P. Henderson, J. Hewitt, D. E. Ho, J. Hong, K. Hsu, J. Huang, T. Icard, S. Jain, D. Jurafsky, P. Kalluri, S. Karamcheti, G. Keeling, F. Khani, O. Khattab, P. W. Koh, M. Krass, R. Krishna, R. Kudipudi, A. Kumar, F. Ladhak, M. Lee, T. Lee, J. Leskovec, I. Levent, X. L. Li, X. Li, T. Ma, A. Malik, C. D. Manning, S. Mirchandani, E. Mitchell, Z. Muniyikwa, S. Nair, A. Narayan, D. Narayanan, B. Newman, A. Nie, J. C. Nieves, H. Nilforoshan, J. Nyarko, G. Ogut, L. Orr, I. Papadimitriou, J. S. Park, C. Piech, E. Portelance, C. Potts, A. Raghuathan, R. Reich, H. Ren, F. Rong, Y. Roohani, C. Ruiz, J. Ryan, C. Ré, D. Sadigh, S. Sagawa, K. Santhanam, A. Shih, K. Srinivasan, A. Tamkin, R. Taori, A. W. Thomas, F. Tramèr, R. E. Wang, W. Wang, B. Wu, J. Wu, Y. Wu, S. M. Xie, M. Yasunaga, J. You, M. Zaharia, M. Zhang, T. Zhang, X. Zhang, Y. Zhang, L. Zheng, K. Zhou, and P. Liang, "On the opportunities and risks of foundation models," 2021. [Online]. Available: <https://arxiv.org/abs/2108.07258>
 - [25] B. Lim, S. O. Arik, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.
 - [26] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 12, 2021, pp. 11 106–11 115.
 - [27] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 22 419–22 430.
 - [28] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the language of time series," 2024. [Online]. Available: <https://arxiv.org/abs/2403.07815>
 - [29] K. Rasul, A. Ashok, A. R. Williams, H. Ghonia, R. Bhagwatkar, A. Khorasani, M. J. D. Bayazi, G. Adamopoulos, R. Riachi, N. Hassen, M. Biloš, S. Garg, A. Schneider, N. Chapados, A. Drouin, V. Zantedeschi, Y. Nevmyvaka, and I. Rish, "Lag-llama: Towards foundation models for probabilistic time series forecasting," 2024. [Online]. Available: <https://arxiv.org/abs/2310.08278>
 - [30] A. Das, W. Kong, R. Sen, and Y. Zhou, "A decoder-only foundation model for time-series forecasting," 2024, accepted at ICML 2024; model commonly referenced as TimesFM. [Online]. Available: <https://arxiv.org/abs/2310.10688>
 - [31] G. J. de Bondt and Y. Sun, "Enhancing GDP nowcasts with ChatGPT: a novel application of PMI news releases," European Central Bank, Frankfurt am Main, Germany, Working Paper Series 3063, Jun. 2025, eCB Working Paper Series No. 3063; released 2025-06-30. [Online]. Available: <https://www.ecb.europa.eu/pub/pdf/scpwps/ecb.wp3063~f88c1b73fc.en.pdf>
 - [32] B. Kwon, T. Park, F. Perez-Cruz, and P. Rungcharoenkitkul, "Large language models: a primer for economists," *BIS Quarterly Review*, Dec. 2024. [Online]. Available: https://www.bis.org/publ/qrtpdf/r_qt2412b.htm
 - [33] A. Carriero, D. Pettenuzzo, and S. Shekhar, "Macroeconomic forecasting with large language models," 2025. [Online]. Available: <https://arxiv.org/abs/2407.00890>
 - [34] C. Ying, T. Cai, S. Luo, S. Zheng, G. Ke, D. He, Y. Shen, and T.-Y. Liu, "Do transformers really perform badly for graph representation?" in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., vol. 34. Curran Associates, Inc., 2021, pp. 28 877–28 888. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2021/file/f1c1592588411002af340cbaedd6fc33-Paper.pdf
 - [35] P. Veličković, W. Fedus, W. L. Hamilton, P. Liò, Y. Bengio, and R. D. Hjelm, "Deep graph infomax," in *International Conference on Learning Representations (ICLR)*, 2019. [Online]. Available: <https://openreview.net/forum?id=rklz9IAcKQ>
 - [36] Y. You, T. Chen, Y. Sui, T. Chen, Z. Wang, and Y. Shen, "Graph contrastive learning with augmentations," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5812–5823.
 - [37] Z. Hou, X. Liu, Y. Cen, Y. Dong, H. Yang, C. Wang, and J. Tang, "GraphMAE: Self-supervised masked graph autoencoders," in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 594–604.
 - [38] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirrmester, and F. Hutter, "Accurate predictions on small data with a tabular foundation model," *Nature*, vol. 637, no. 8045, pp. 319–326, Jan. 2025, published online 8 Jan 2025. [Online]. Available: <https://www.nature.com/articles/s41586-024-08328-6>
 - [39] D. Ereemeev, G. Bazhenov, O. Platonov, A. Babenko, and L. Prokhorenkova, "Turning tabular foundation models into graph foundation models," 2025. [Online]. Available: <https://arxiv.org/abs/2508.20906>
 - [40] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=WE_vluYUL-X
 - [41] H. Yang, X.-Y. Liu, and C. D. Wang, "FinGPT: Open-source financial large language models," 2023. [Online]. Available: <https://arxiv.org/abs/2306.06031>
 - [42] Y. Luo, Y. Feng, J. Xu, P. Tascas, and Y. Liu, "Llm-powered multi-agent system for automated crypto portfolio management," *arXiv preprint arXiv:2501.00826*, 2025.
 - [43] J. Jeon and H. Lee, "Can blindfolded LLMs still trade? An anonymization-first framework for portfolio optimization," 2026. [Online]. Available: <https://arxiv.org/abs/2603.17692>
 - [44] X. Hu, W. Y. Chan, Y. Shi, Q. Sun, W.-C. Wang, C. Wu, H. Wang, and N. He, "An effective and cost-efficient agentic framework for ethereum smart contract auditing," 2026. [Online]. Available: <https://arxiv.org/abs/2601.17833>
 - [45] Z. Kong, W. Xia, C. Wang, Y. Lu, P. Li, S. Li, Z. Cao, and Y. Liu, "Knowdit: Agentic smart contract vulnerability detection with auditing knowledge summarization," 2026. [Online]. Available: <https://arxiv.org/abs/2603.26270>
 - [46] L. Liu, W. Zhang, L. Wei, H. Guan, Y. Tian, Y. Liu, and S.-C. Cheung, "LLM-powered detection of price manipulation in DeFi," 2025. [Online]. Available: <https://arxiv.org/abs/2510.21272>
 - [47] A. Watson, G. Richards, and D. Schiff, "Explain first, trust later: LLM-augmented explanations for graph-based crypto anomaly detection," *arXiv preprint arXiv:2506.14933*, 2025.
 - [48] W. Hu, M. Fey, M. Zitnik, Y. Dong, H. Ren, B. Liu, M. Catasta, and J. Leskovec, "Open graph benchmark: Datasets for machine learning on graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
 - [49] C. Bergmeir and J. M. Benítez, "On the use of cross-validation for time series predictor evaluation," *Information Sciences*, vol. 191, pp. 192–213, 2012, data Mining for Software Trustworthiness. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0020025511006773>
 - [50] P. Liang, R. Bommasani, T. Lee, D. Tsipras, D. Soylu, M. Yasunaga, Y. Zhang, D. Narayanan, Y. Wu, A. Kumar *et al.*, "Holistic evaluation of language models," *arXiv preprint arXiv:2211.09110*, 2022.
 - [51] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan, "Swe-bench: Can language models resolve real-world github issues?" in *International Conference on Learning Representations (ICLR)*, 2024.

- [52] X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang, S. Zhang, X. Deng, A. Zeng, Z. Du, C. Zhang, S. Shen, T. Zhang, Y. Su, H. Sun, M. Huang, Y. Dong, and J. Tang, "Agentbench: Evaluating llms as agents," in *International Conference on Learning Representations (ICLR)*, 2024.
- [53] DefiLlama, "DefiLlama: DeFi TVL aggregator and analytics," 2026, data used: 2025; accessed: 2026-02-02. [Online]. Available: <https://defillama.com>
- [54] R. A. Shahzad Qaiser, "Text mining: Use of tf-idf to examine the relevance of words to documents," *International Journal of Computer Applications*, vol. 181, no. 1, pp. 25–29, Jul 2018. [Online]. Available: <https://ijcaonline.org/archives/volume181/number1/29681-2018917395/>

APPENDIX A

FULL ALL-HORIZON `b1_forecast` RESULTS

Table 12 reports the complete `b1_forecast` metrics across all four horizons $h \in \{1, 4, 8, 12\}$ and all four prediction methods. The main text (§6) summarises the cross-metric pattern; this appendix provides the full per-horizon audit for readers reproducing the leaderboard.

TABLE 12

`b1_forecast` all-horizon forecasting on the 2025 test split. PageRank MAE shown in 10^{-5} units. Best per horizon in **bold**. Trend consistency for `m1_persistence` is structurally 0 because $\hat{G}_{t+h} = G_t$ produces no trend signal.

h	Method	PR MAE ($\downarrow$)	HHI MAE ($\downarrow$)	Gini MAE ($\downarrow$)	Rank Corr ($\uparrow$)	Trend Cons ($\uparrow$)
1	<code>m1_persistence</code>	3.6	0.0376	0.0027	0.556	0.000
	<code>m3_evolvegcn</code>	3.7	0.0481	0.1149	0.535	0.444
	<code>m4_fm_only</code>	4.7	0.0376	0.0029	0.549	0.531
	<code>m5_fm_rules</code>	4.7	0.0376	0.0029	0.549	0.525
4	<code>m1_persistence</code>	3.4	0.0322	0.0018	0.570	0.000
	<code>m3_evolvegcn</code>	3.5	0.0489	0.1326	0.551	0.324
	<code>m4_fm_only</code>	4.5	0.0321	0.0022	0.559	0.621
	<code>m5_fm_rules</code>	4.5	0.0321	0.0022	0.558	0.628
8	<code>m1_persistence</code>	3.9	0.0399	0.0028	0.517	0.000
	<code>m3_evolvegcn</code>	4.0	0.0492	0.1140	0.499	0.296
	<code>m4_fm_only</code>	4.5	0.0398	0.0030	0.508	0.624
	<code>m5_fm_rules</code>	4.5	0.0398	0.0030	0.508	0.632
12	<code>m1_persistence</code>	3.7	0.0393	0.0027	0.525	0.000
	<code>m3_evolvegcn</code>	3.8	0.0534	0.1245	0.509	0.314
	<code>m4_fm_only</code>	4.1	0.0392	0.0029	0.516	0.600
	<code>m5_fm_rules</code>	4.1	0.0392	0.0029	0.516	0.600

APPENDIX B

PER-SCENARIO $\mathcal{B}_4$ STRESS RESULTS

Table 13 reports the per-scenario stress-test outputs across all five standardised scenarios $\mathcal{S}_0$ and all four methods. The main text aggregates these into a single qualitative finding (§6); this appendix retains the full audit artifact.

TABLE 13

$\mathcal{B}_4$ stress per-scenario detail. Loss MAE in normalised units; Distress count MAE in protocols; target overlap@10 measured against the actual future graph. Best per scenario row in **bold**. The main pattern is conservative: FM variants roughly match the strong persistence baseline on loss and target-overlap metrics, while EvolveGCN is generally weaker on downstream what-if fidelity.

Scenario	Method	Loss MAE ($\downarrow$)	Distress MAE ($\downarrow$)	Prop. Depth MAE ($\downarrow$)	Overlap@10 ($\uparrow$)
S1 (single protocol)	m1_persistence	0.0668	200.3	0.897	0.314
	m3_evolvegcgn	0.1072	218.1	0.966	0.034
	m4_fm_only	0.0667	201.4	0.897	0.314
	m5_fm_rules	0.0667	201.6	0.897	0.314
S2 (cross-protocol)	m1_persistence	0.0150	27.3	0.000	0.521
	m3_evolvegcgn	0.0116	196.4	0.000	0.189
	m4_fm_only	0.0151	50.3	0.000	0.525
	m5_fm_rules	0.0151	52.1	0.000	0.525
S3 (bridge)	m1_persistence	0.0079	0.0	0.000	0.445
	m3_evolvegcgn	0.0156	0.0	0.000	0.269
	m4_fm_only	0.0080	0.0	0.000	0.445
	m5_fm_rules	0.0080	0.0	0.000	0.445
S4 (stablecoin)	m1_persistence	0.0100	0.0	0.034	0.536
	m3_evolvegcgn	0.0200	0.0	0.034	0.293
	m4_fm_only	0.0099	0.0	0.034	0.536
	m5_fm_rules	0.0099	0.0	0.034	0.536
S5 (lending)	m1_persistence	0.0100	0.0	0.000	0.602
	m3_evolvegcgn	0.0538	0.0	1.000	0.330
	m4_fm_only	0.0099	0.0	0.000	0.602
	m5_fm_rules	0.0099	0.0	0.000	0.602

APPENDIX C

PER-REGIME b_6 ROBUSTNESS RESULTS

Table 14 expands the single mean relative-degradation figure reported in Table 8 into the full 5×4 matrix over the five degradation regimes and four prediction methods. The FM’s robustness claim in §6 (“24% less average degradation than persistence”) is the column average of the Δ_{rel} column; the per-regime view here lets readers see which regimes drive that gap (low_data_10pct and partial_graph_30) versus which are essentially ties (noisy_features_01, missing_features_20, where persistence is structurally invariant).

TABLE 14

b_6 _robustness per-regime detail. All metrics at $h=4$ on the 2025 test split under five degradation regimes. PageRank, HHI, and Gini MAE in 10^{-3} units. Relative degradation is signed against the clean b_1 _forecast baseline; negative values mean the predictor is no worse than on clean data. Best per metric (lowest MAE, highest rank correlation, smallest $|\Delta_{\text{rel}}|$) in **bold**. Persistence ignores node features, so its noisy_features_01 and missing_features_20 rows degenerate to the clean numbers (and $\Delta_{\text{rel}} = 0$).

Regime	Method	PR MAE ($\downarrow$)	HHI MAE ($\downarrow$)	Gini MAE ($\downarrow$)	Rank Corr ($\uparrow$)	Δ_{rel} (~ 0)
low_data_10pct	m1_persistence	15.94	0.0591	0.0023	0.552	+0.766
	m3_evolvegcgn	15.23	0.0999	0.1301	0.535	+0.276
	m4_fm_only	12.83	0.0586	0.0027	0.528	+0.612
	m5_fm_rules	12.71	0.0587	0.0027	0.528	+ 0.607
low_data_25pct	m1_persistence	9.46	0.0282	0.0020	0.577	−0.093
	m3_evolvegcgn	10.22	0.0486	0.1281	0.558	− 0.027
	m4_fm_only	9.86	0.0279	0.0024	0.553	−0.126
	m5_fm_rules	9.95	0.0279	0.0024	0.553	−0.126
partial_graph_30	m1_persistence	9.63	0.0350	0.0019	0.554	+0.068
	m3_evolvegcgn	10.36	0.0488	0.1236	0.537	− 0.048
	m4_fm_only	12.52	0.0350	0.0022	0.500	+0.082
	m5_fm_rules	12.45	0.0350	0.0022	0.499	+0.077
noisy_features_01	m1_persistence	9.77	0.0322	0.0018	0.584	0.000
	m3_evolvegcgn	10.68	0.0489	0.1326	0.563	0.000
	m4_fm_only	11.38	0.0323	0.0021	0.554	−0.005
	m5_fm_rules	11.35	0.0323	0.0021	0.554	−0.007
missing_features_20	m1_persistence	9.77	0.0322	0.0018	0.584	0.000
	m3_evolvegcgn	10.68	0.0489	0.1326	0.563	0.000
	m4_fm_only	12.11	0.0323	0.0023	0.559	+0.017
	m5_fm_rules	12.08	0.0323	0.0023	0.559	+0.014

APPENDIX D

PER-BUDGET `b2_warning` RESULTS

Table 15 reports per-event precision, recall, F1-warning, and alert stability across alert budgets $K \in \{5, 10, 20\}$ for the three stress events used in §6. The main text and Table 10 focus on lead time and the qualitative timeline; this table records the precision/recall trade-off that the budget parameter actually controls and is provided for readers who wish to reproduce or recalibrate the `b2_warning` leaderboard against alternative monitors.

TABLE 15

`b2_warning` per-event detail for the shared weighted-degree monitor (`hl_weighted_degree`) across alert budgets $K \in \{5, 10, 20\}$. Recall and F1-warning shown in 10^{-3} units (the full DeFi corpus contains $\sim 4,300$ protocols, so per-week recall denominators are large). Lead time is event-level and not budget-sensitive: 5 weeks for Terra/Luna and FTX, 4 weeks for SVB-USDC. `b2_warning` sits upstream of the predictor choice, so a single monitor is shared across all methods (see §6).

Event	K	Prec. ($\uparrow$)	Rec. $\times 10^3$ ($\uparrow$)	F1 _w $\times 10^3$ ($\uparrow$)	Stab. ($\uparrow$)	$ \mathcal{A} $
Terra/Luna	5	0.60	1.81	3.61	0.44	11
	10	0.70	3.26	6.49	0.62	19
	20	0.65	7.97	15.74	0.74	36
FTX	5	0.60	1.33	2.65	0.64	10
	10	0.70	2.39	4.77	0.72	18
	20	0.55	4.79	9.49	0.72	39
SVB-USDC	5	1.00	1.81	3.62	0.55	10
	10	1.00	2.82	5.62	0.75	15
	20	1.00	5.64	11.22	0.81	30

APPENDIX E

PER-REGIME A1 (DATA-HEALTH GATE) ISOLATION

Table 16 expands Panel B of Table 11 into the per-regime audit underlying the “FIR $\in [0.27, 0.60]$ ” range reported in the main text. The four regimes vary which degradation channel is active (*topo* only, *feat* only, *both*, and a milder *severe* combination) and the three rows per regime sweep the data-health threshold τ_{data} , holding the confidence gate disabled ($\tau_{\text{conf}}=0$) so that A1 is the only suppression mechanism. The structurally important pattern is that *strict* ($\tau_{\text{data}}=0.85$) drives the safe-mode fraction to 100% in every regime and collapses FIR to 0 by suppressing intervention tickets, while *off* produces FIR up to 0.541.

TABLE 16

Per-regime data-health gate ablation isolating A1 from A2. Each row holds $\tau_{\text{conf}}=0$ and sweeps τ_{data} . “*strict*” = $\tau_{\text{data}}=0.85$, “*on*” = 0.70, “*off*” = 0.00. The four regimes apply targeted edge or feature degradation on the clean 2025 test split ($n_{\text{weeks}}=29$): *extreme* masks both topology and features at 90%, *topo* removes 90% of edges only, *feat* corrupts 90% of node features only, *severe* masks both at 80%. Numbers expand Panel B of Table 11: under degradation the gate-off setting produces FIR up to 0.541, while *on/strict* can suppress intervention tickets via safe-mode and drive FIR to 0.

Regime	Gate	τ_{data}	$\overline{\text{dh}}$	Safe-mode (%)	Ticket Prec. ($\uparrow$)	FIR ($\downarrow$)	Interventions
extreme	strict	0.85	0.587	100	0.510	0.000	0
	on	0.70	0.587	100	0.545	0.000	0
	off	0.00	0.587	0	0.490	0.510	57
extreme_topo	strict	0.85	0.637	100	0.517	0.000	0
	on	0.70	0.637	100	0.462	0.000	0
	off	0.00	0.637	0	0.559	0.441	58
extreme_feat	strict	0.85	0.755	100	0.503	0.000	0
	on	0.70	0.755	0	0.510	0.267	5
	off	0.00	0.755	0	0.503	0.267	5
severe	strict	0.85	0.751	100	0.490	0.000	0
	on	0.70	0.751	0	0.538	0.400	25
	off	0.00	0.751	0	0.538	0.541	24



PLACE
PHOTO
HERE

Aijie Shu



PLACE
PHOTO
HERE

Fengxiang He is a lecturer with the School of Informatics, University of Edinburgh. His research interest is in trustworthy AI, particularly deep learning theory, decentralised learning, privacy and fairness in machine learning, symmetry in machine learning, machine learning for game-theoretical problems, and their applications in economics and finance. He is an area chair of ICML, NeurIPS, ICLR, UAI, AISTATS, and ACML, and an associate editor of IEEE Transactions on Technology and Society.